To compute the probability of a process, one must sum up over all the possible spinfoams within the box which have the same boundary as that process. The boundary of a spinfoam is a spin network and the matter on it.

The equations of loop quantum gravity express the probability of a process in terms of sums over spinfoams with given boundaries. In this way it is possible to compute, in principle, the probability of any physical event.[fn42]

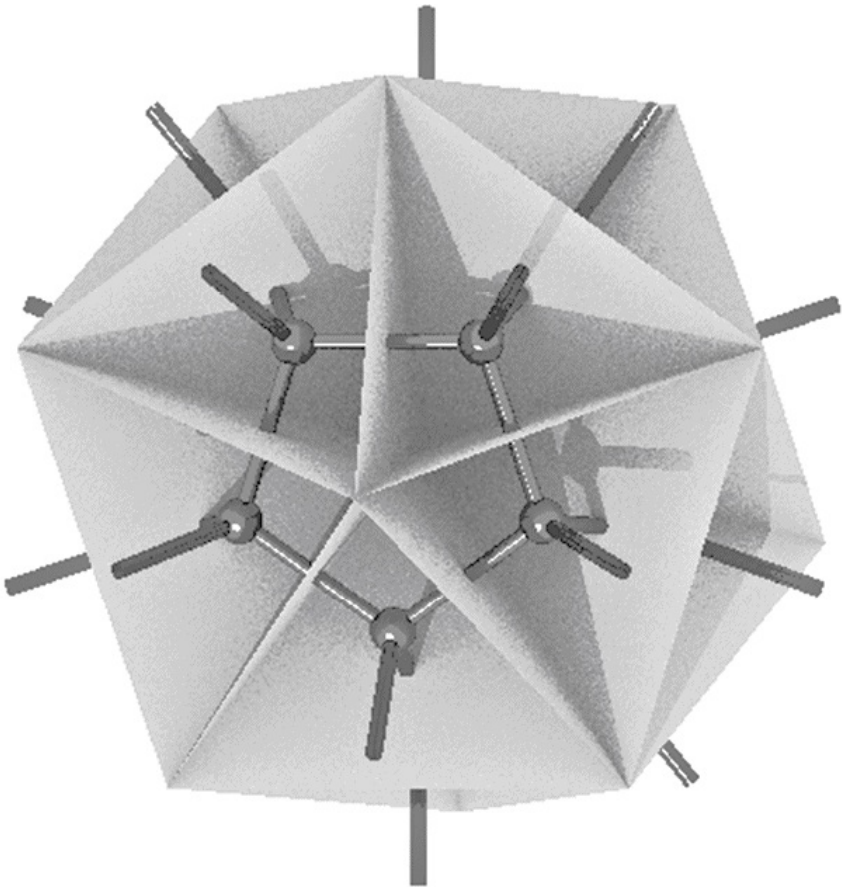


Figure 7.4 A vertex of spinfoam. Courtesy of Greg Egan.

At first sight, this way for making calculations in quantum gravity, based on spinfoams, seems very different from the usual ways in which things are computed in theoretical physics. There is no given space, no given time, and spinfoams seem objects quite remote from, say, the particles of the standard model. But in fact there are strong similarities between the spinfoam technique and the calculation techniques used in the standard model. In fact, even more than this, the spinfoam technique is actually a beautiful merging of the two main calculation techniques used in the context of the standard model: Feynman diagrams and the lattice approximation.

Feynman diagrams are used, for instance, to compute processes dominated by electromagnetic or weak forces. A Feynman diagram represents a sequence of elementary interactions among particles. An example is in figure 7.5,